

Economic History II

Bison, reservations and allotment

FGV EPGE

Outline of the class

- **Feir, Gillezeau and Jones (2024)**: the near extinction of the bison and the reversal of fortunes on the Great Plains.
- **Dippel (2014)**: reservations that combined autonomous bands, and their incomes today.
- **Dippel, Frye and Leonard (2024)**: which BIA agent was posted where, and what happened to the land.

The slaughter of the bison

Feir, Gillezeau and Jones (2024)

In the late nineteenth century, the North American bison was brought to the brink of extinction in less than two decades. We demonstrate that the loss of the bison had immediate, negative consequences for the Native Americans who relied on them and ultimately resulted in a persistent reversal of fortunes. Once amongst the tallest people in the world, the generations of bison-reliant people born after the slaughter lost their entire height advantage. By the early twentieth century, child mortality was 16 percentage points higher and the probability of reporting an occupation 19 percentage points lower in bison nations compared with nations that were never reliant on the bison. Throughout the latter half of the twentieth century and into the present, income per capita has remained 25% lower, on average, for bison nations. This persistent gap cannot be explained by differences in agricultural productivity, self-governance, or application of the *Dawes Act*. We provide evidence that this historical shock altered the dynamic path of development for formerly bison-reliant nations. We demonstrate that limited access to credit constrained the ability of bison nations to adjust through re-specialization and migration.

Key words: North American bison, Buffalo, Extinction, Economic history, Native Americans, Indigenous, Income shock, Intergenerational mobility

JEL codes: I15, J15, N31, N32, O10

“But when the buffalo went away the hearts of my people fell to the ground, and they could not lift them up again.” (Chief Crow Plenty-Coups quoted in [Lindermann \(1930, p. 169\)](#))

A reversal of fortunes

- Before the 1870s, the bison-reliant nations were among the wealthiest peoples in North America.
- Today the Native communities of the Great Plains have some of the lowest incomes in the United States.
- Between the two: the loss of one resource, in under two decades.
- The paper asks whether that single shock accounts for the **reversal of fortunes**.

“But when the buffalo went away the hearts of my people fell to the ground, and they could not lift them up again.”

Chief Crow Plenty-Coups, quoted in Lindermann (1930), epigraph of the paper

The bison economy

- Hunted on the Plains for at least **10,000 years** before European contact.
- Not only food: hides for clothing, lodging and blankets, bones for tools, marrow eaten, stomachs made into bags.
- Pemmican, dried meat with berries and fat, kept for years inside a bison stomach. Storable insurance against shortage.
- Well-defined ownership over hunting grounds, permanent sites of residence, complex kinship networks.
- Anthropometric evidence: living standards comparable to or better than their European contemporaries.

The collapse

- Before European settlement, between 10 and 30 million bison, from the Rockies to the Appalachians.
- Stock at the start of the rapid slaughter: about 8 million. At the end: fewer than 500.
- 1871: tanners in England and Germany find a commercially viable way of tanning bison hides.
- 1875: 1 million hides shipped from the United States to France and England alone.
- Southern herd gone by 1879. The last bison in the remaining territory, 1883.

Treaty rights and the army

- The treaties of the late 1860s traded large tracts of ancestral territory for annuities and for the exclusive right to hunt the herds.
- Many included clauses protecting the bison from settler hunting. Those clauses were not enforced.
- American military commanders believed Native nations would not settle onto reservations while the bison lived.
- Generals encouraged troops to kill bison for food, sport, or “practice”.
- Gwynne: a country once “black and brown with bison was left white by bones bleaching in the sun”.

Limits on adjustment

- Leaving a reservation required permission from a federal Indian agent, and departures were discouraged.
- Free movement came with the Indian Citizenship Act of 1924, and only for those born after 1924.
- Cattle ranching, the obvious alternative use of the same skills and land, was blocked by agents or by lack of credit.
- Agriculture was effectively the only economic activity the federal government supported.

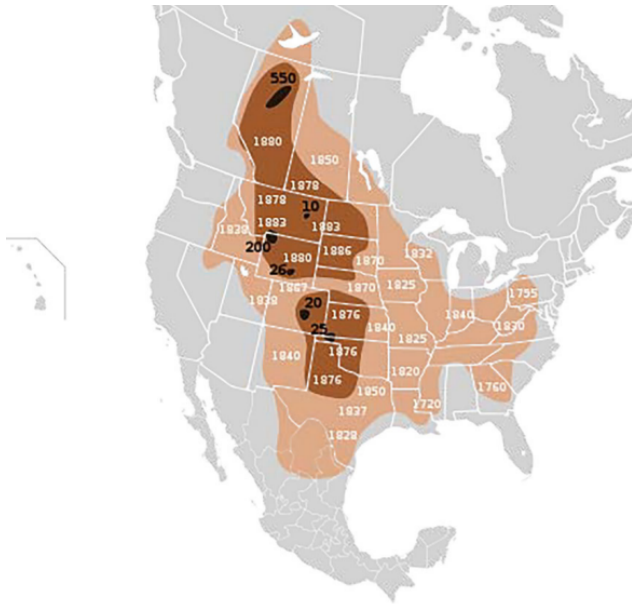


FIGURE 1
The North American bison's range in 1730, 1870, and 1889



Hornaday's map

- William Hornaday was commissioned by the Smithsonian to document the bison and its elimination.
- Outer shaded region: the range as of 1730. Middle region: the range in 1870, at the start of the rapid slaughter.
- The black patches are the herds left in 1889, all of them in ranches in captivity, with their sizes labeled.
- This is where the geography of the **rapid slaughter** comes from.

Measuring bison reliance

- Overlay Hornaday's map with maps of ancestral territory: which nations had land inside the slaughter region.
- Geography alone misclassifies. The Mandan lived in bison-dense North Dakota but farmed, and traded for bison meat.
- Second condition: bison listed as a primary food source in the 15 volumes of the Smithsonian *Handbook of North American Indians*.
- Treatment is both conditions together. The comparison group is nations that were never bison-reliant.



Notes: This figure displays a digital reproduction of detailed regional maps of traditional Indigenous territories from the “General Ethnic Map of Native North America ca A.D. 1600” from Martin and O’Leary (1990).

Data

- Height: nearly 9,000 Native Americans measured by **Franz Boas** between 1889 and 1903.
- 1900 census over-samples: child mortality, children ever born, sex composition, occupations.
- Income per capita of Native Americans living inside reservation boundaries: 1945, then 1970 through 2019.
- Mechanisms: banks, population, manufacturing, industry of employment, migration.

TABLE 1
Summary statistics for pre-1900 outcomes and controls by exposure to the rapid slaughter

	Exposed to Slaughter (1)	Not Bison-Reliant (2)	Difference (3)
<i>Panel A: Nation Level Characteristics</i>			
EA Fully/Partly nomadic	0.609 (0.499)	0.245 (0.432)	-0.363**
EA Historic Centralization	0.565 (0.507)	0.500 (0.502)	-0.065
EA Wealth Differences	0.130 (0.344)	0.340 (0.476)	0.209*
Log Population as of 1500	6.111 (1.645)	5.005 (2.615)	-1.105*
Log Ruggedness	2.756 (0.885)	3.331 (1.105)	0.575*
Observations	23	106	129
<i>Panel B: Franz Boas Data</i>			
Standing Height (CM)	156.58 (18.24)	150.56 (20.64)	-6.02***
Male Standing Height (CM) pre-slaughter	171.25 (6.42)	168.62 (6.56)	-2.63***
Female Standing Height (CM) pre-slaughter	158.82 (5.35)	155.95 (5.57)	-2.87***
Female	0.34 (0.47)	0.41 (0.49)	0.07***
Observations	2753	6238	8991
<i>Panel C: IPUMS 1900 Historical Over-Samples—Women</i>			
% of Children Surviving	0.76 (0.25)	0.83 (0.22)	0.07***
Children ever born	4.93 (2.61)	4.68 (2.47)	-0.25**
Observations	1366	2412	3778
<i>Panel D: IPUMS 1900 Historical Over-Samples—Men</i>			
Std Occupational Score	0.03 (0.75)	0.49 (0.64)	0.45***
Has an Occupation	0.52 (0.40)	0.87 (0.33)	0.35***
In Agriculture	0.42 (0.49)	0.59 (0.49)	0.17***
Observations	2439	4411	6850

Table 1: before and after

- 23 exposed nations against 106 that were never bison-reliant.
- Exposed nations were more often fully or semi-nomadic, with fewer wealth distinctions and larger populations as of 1500.
- Before the slaughter, bison-reliant men were 2.6 cm taller and women 2.9 cm taller than the comparison group.
- By 1900 the ranking has flipped: fewer surviving children, lower occupational rank, 35 percentage points less likely to report an occupation.

Heights: the comparison

- Boas records nation, sex, age and height, so every person can be placed in a birth cohort.
- Compare people born before and after the bison were gone, in bison nations and in nations that never relied on them.
- The bison were effectively gone by **1886**: the last herds died by 1883, and pemmican kept for about three more years.
- Held fixed: the nation itself, age, survey year, wars with the United States, railways, land cessions, settler density.

Table 2: heights

- After 1886, heights in exposed nations fall 2 to 3 cm relative to the comparison group.
- Main estimate: **2.62 cm**. The advantage before the slaughter had been about 2.7 cm.
- In height-for-age terms, a fall of about a third of a standard deviation.
- Each additional growing year without the bison, between ages 0 and 21, costs about 0.45 cm.

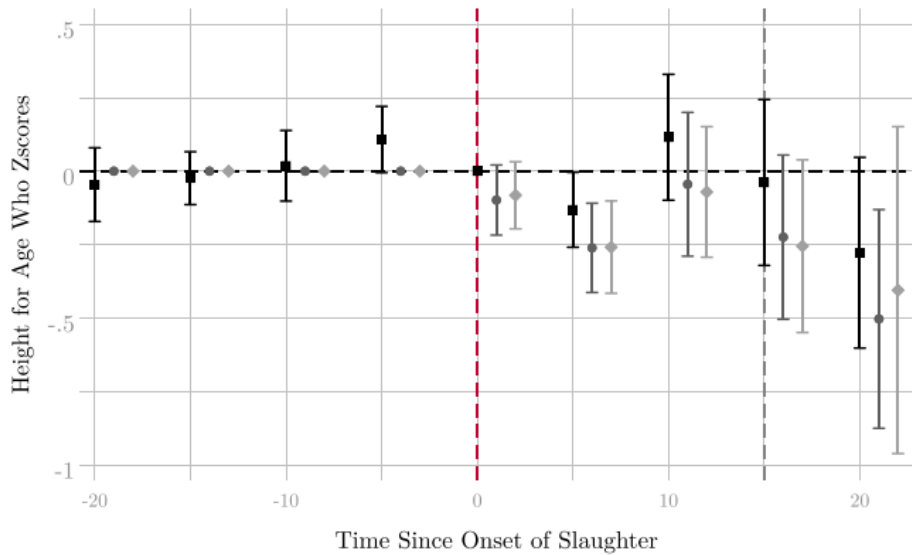


Figure 2: cohort by cohort

- Horizontal axis: years since the slaughter began in 1871. The second dashed line marks 1886.
- Cohorts born before 1871 show no gap, so the two groups were on the same path where that can be checked.
- Cohorts born after the start show a widening fall in height for age.
- Three versions: baseline, one allowing bison nations their own prior trend, one adding colonial, cultural and geographic controls.

Two caveats about Boas

- Survivorship. If the shortest children were the ones who died, the surviving heights understate the damage.
- Selection. Work on the Cherokee suggests Boas's sample is representative on average but over-represents the top and the bottom.
- What the design needs is that this over- or under-representation does not differ across birth cohorts and across the two groups.
- The 1900 census provides an independent check on the same period.

Figure A3: Event Study Coefficients Indicating the Relative Declines in Height of those Exposed to the Rapid Slaughter.

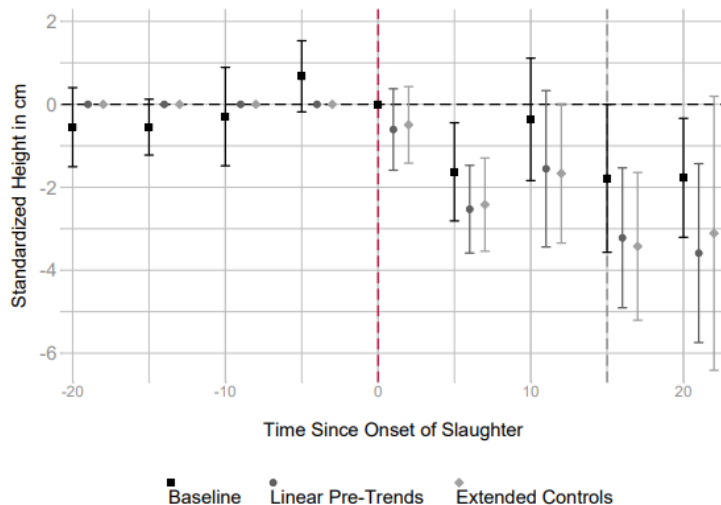
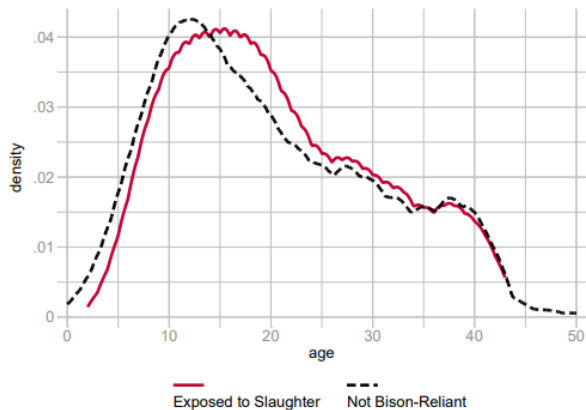
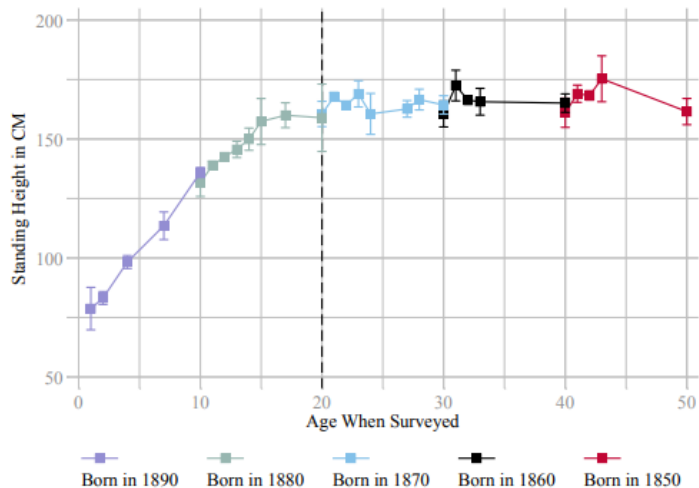


Figure A4: The Age Distributions of Individual Heights by Exposure to the Slaughter



as: This figure displays the age distribution of individuals who were from nations that were exposed to the ghter and those that were not bison-reliant. The sample sizes of individuals exposed to the slaughter and e that were not bison-reliant are $N=2,753$ and $N=6,238$, respectively. The data are from Franz Boas' sample ividual heights, collected between 1890 and 1901.

Figure A5: Heights by Age and Birth Cohort



Appendix figures A3 to A5

- A3: the same cohort pattern with height in centimeters instead of height for age.
- A4: bison nations have visibly fewer children in Boas's sample, consistent with higher child mortality.
- A5: within a birth cohort, height does not depend on the year Boas happened to survey. Little sign of age-related selection.

1900: mortality and sex ratios

- Women exposed to the slaughter had 10 percentage points fewer of their children survive.
- Among mothers born after the slaughter began, 15.5 percentage points.
- No offsetting rise in the number of children ever born.
- Cohorts born after 1871 are about 6 percentage points more likely to be female, after 1886 about 9. Male fetuses and newborns are the more fragile, so this reads as **maternal deprivation**.

1900: occupations

- Working-age men from bison nations were **19 percentage points** less likely to report any occupation.
- The same holds for men born after 1871, who never worked in a bison economy.
- Among men who did report an occupation, the rank is no different. The gap comes from reporting none at all.
- Enumerators were not leaving the field blank more often for bison nations. The answers were non-occupation responses.

Bounding the survivorship problem

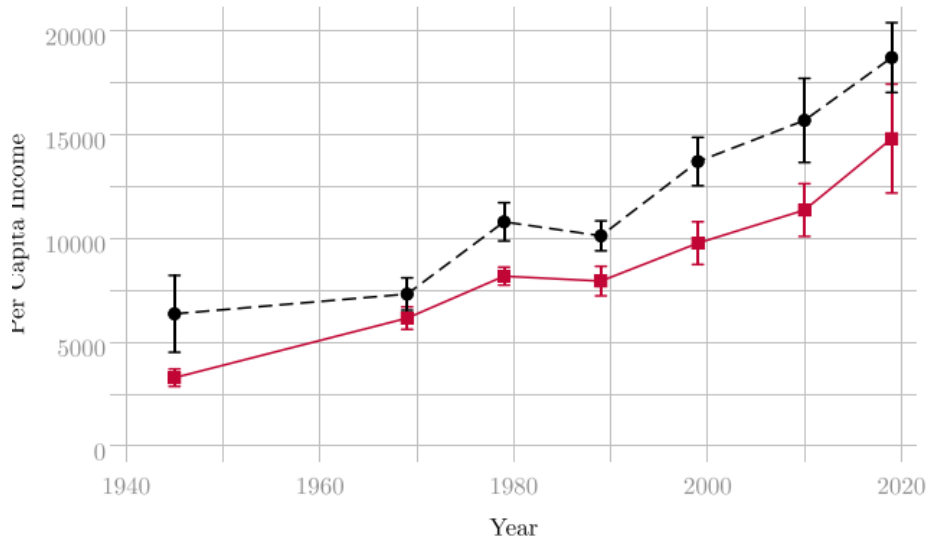
- Take child mortality of 15 percent, the top of the range estimated from the 1900 data.
- Assume the children who died would have stood at the 20th percentile of the surviving height distribution.
- The 2.62 cm estimate then lies between 3.5 cm and 2.02 cm.
- The sign and the order of magnitude survive the assumption either way.

Did they recover?

1945 to 2019

The persistence question

- Large regional shocks are normally absorbed. Labor and capital move, regions re-specialize.
- Unit of observation: the reservation, or the nation associated with it.
- Outcome: income per capita of Native Americans living within reservation boundaries, in 2010 dollars.
- Coverage: 1945, 1970, 1980, 1990, 2000, 2007 to 2012, and 2015 to 2019.



Exposed to Slaughter



Not Bison Reliant

Figure 3: income

- Exposed nations sit below the comparison group in every period from 1945 to 2019.
- The two lines do not close.
- Seven decades is long enough for the adjustment to have happened.

Table 4: the size of the gap

- Income on formerly bison-reliant reservations is about **25 percent** lower across the whole period.
- In 2010 dollars, roughly \$3,100 a year.
- Mean income per capita on reservations in the sample: \$11,781.
- So the gap is about a quarter of an income that is already among the lowest in the country.

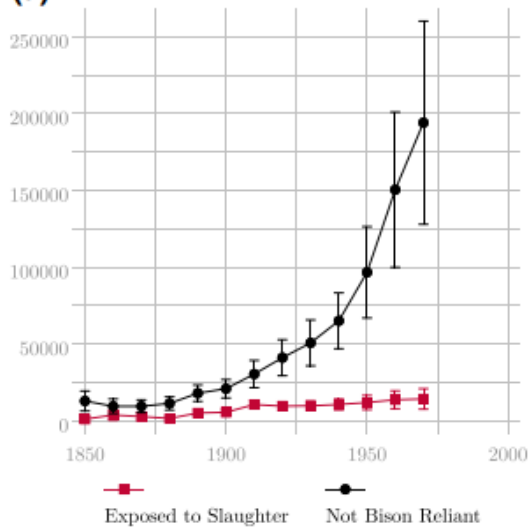
Table 4: what does not explain it

- Geography: ruggedness, area, latitude.
- Culture: nomadism, wealth distinctions, population density in 1500 and 1600.
- Colonial history: forced coexistence, historical centralization, railways, land cessions, settler density, battles, distance displaced.
- Later policy and shocks: the Dust Bowl, crop and livestock values, prime land, the Dawes Act and fractionated ownership, the IRA and self-governance compacts.

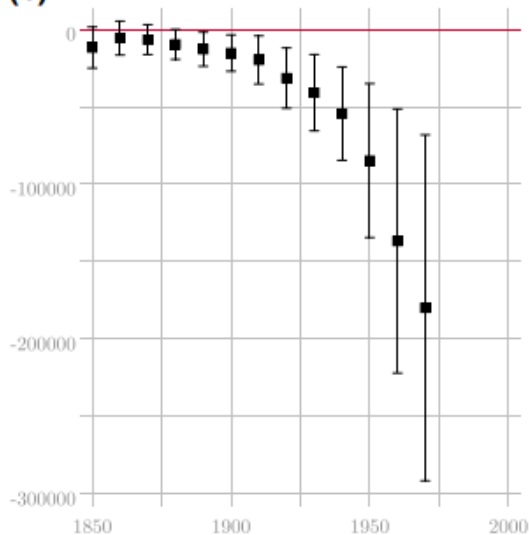
Why it persisted

Why factors did not adjust

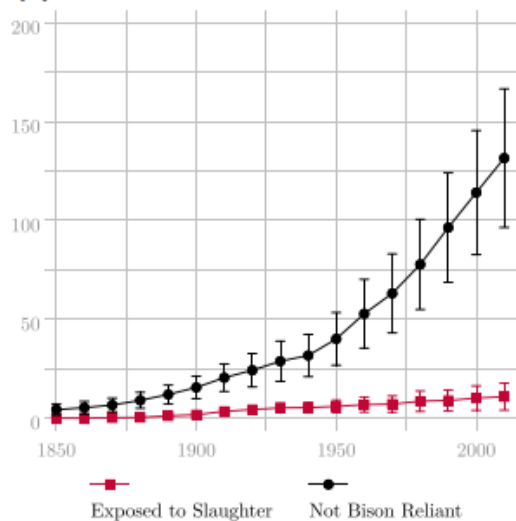
- When a region loses its industry, people and capital normally move until returns even out.
- Both margins are costly, and both need money up front.
- The claim: the loss of the bison made these regions unattractive relative to similar places, and thin **credit** kept them that way.
- Evidence in four parts: population, banks, industry, migration.

(a)

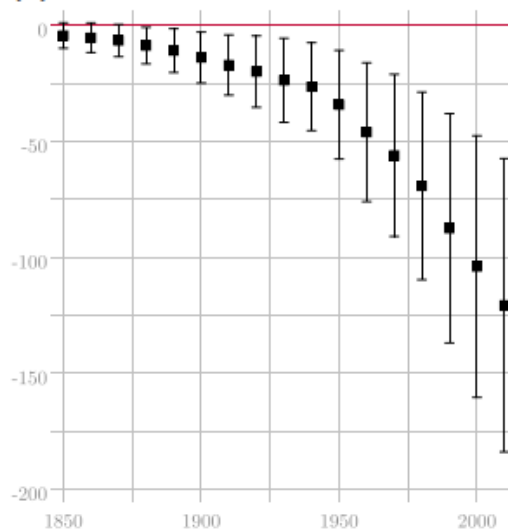
White Population

(b)

White Population

(c)

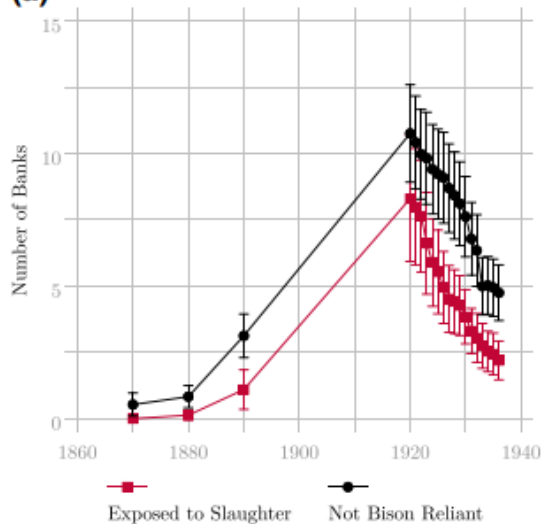
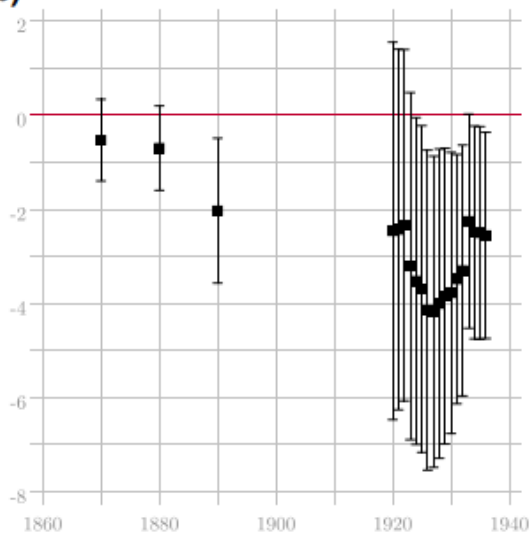
Population Density

(d)

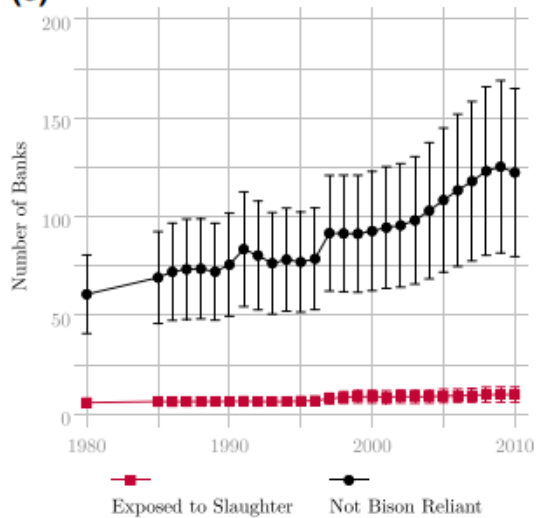
Population Density

Figure 4: population

- Until 1890, white population and population density track each other across the two sets of regions.
- After the turn of the twentieth century they separate.
- Settlers kept arriving where there was economic opportunity, and increasingly that was not the former bison range.

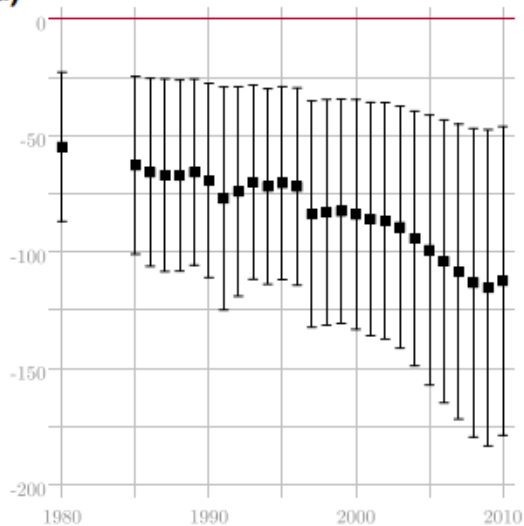
(a)**(b)**

(c)



Number of Banks, Post-Depression

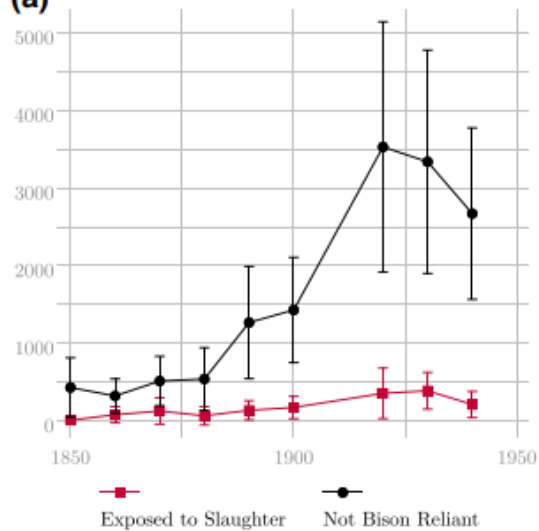
(d)



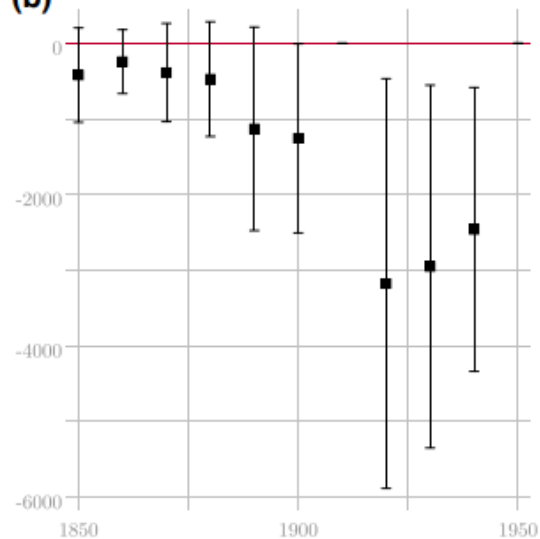
Number of Banks, Post-Depression

Figure 5: banks

- Number of banks in the counties overlapping each reservation, 1870 to 1936 and 1980 to 2010.
- Financial development is slower in the former bison range at both ends of the century.
- Unequal access to credit within the Indigenous population is not a modern phenomenon.
- Distance to the nearest bank in **1870** becomes the paper's measure of early credit access.

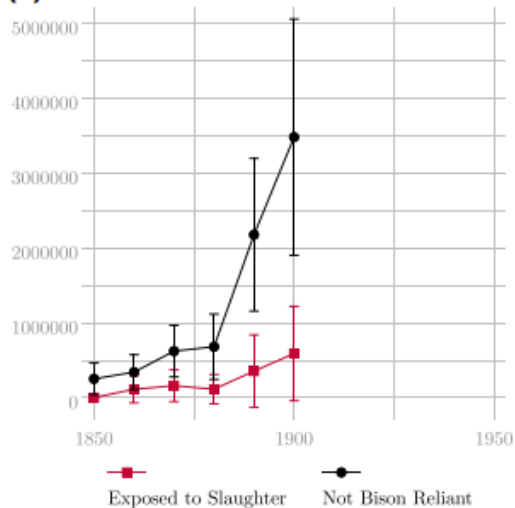
(a)

Employment

(b)

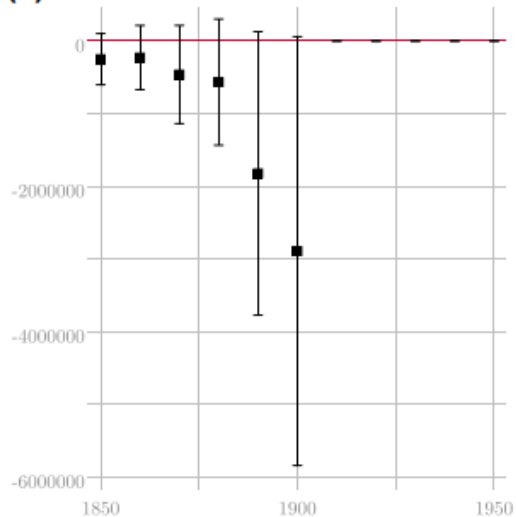
Employment

(c)



Capital Invested

(d)



Capital Invested

Figure 6: manufacturing

- Before the slaughter, manufacturing employment and capital invested are essentially equal across the two groups.
- Afterwards they diverge.
- The United States was moving out of agriculture and deepening its capital stock across every major sector.
- The former bison range did not take part in that.

Table 5: industry of employment

- In 1900, men from bison nations were 30 percentage points less likely to report an industry. By 1930, 10 points.
- By 1930 the gap in agriculture had closed completely.
- The gap in the other capital-intensive industries, manufacturing, mining and construction, had widened.
- Agriculture was the sector the BIA financed. The others needed ordinary credit.

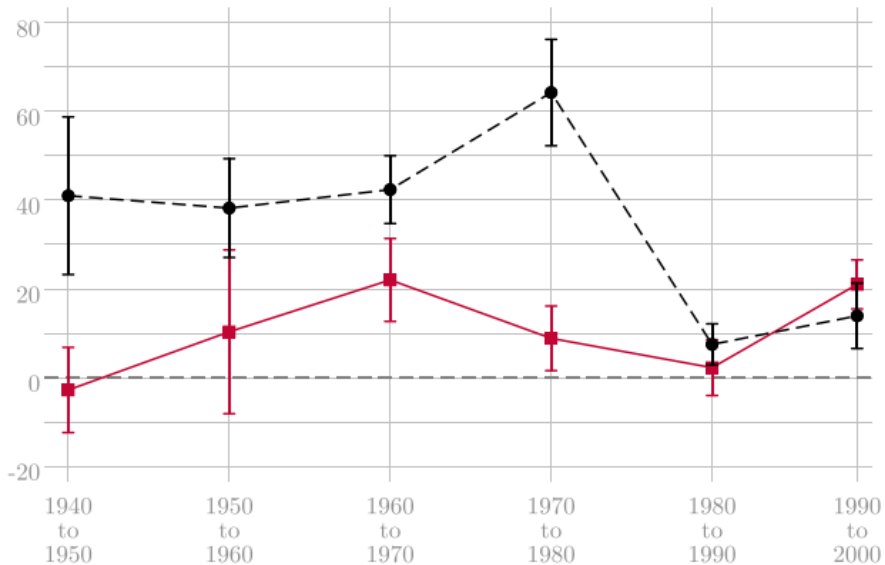
Table 5: distance to a bank in 1870

- Among men who reported an industry, those from bison nations farther from a bank in 1870 were more likely to be farming by 1930.
- The same men were less likely to be in manufacturing, mining or construction.
- Distance to the nearest metropolitan area is held fixed, so the comparison is between equally isolated places with different access to credit.
- BIA agricultural credit substituted for ordinary credit, and held nations in the sector the rest of the country was leaving.

Income sources, 1915 to 1920

- BIA annual reports give income and wealth by nation for 52 nations.
- Formerly bison-reliant nations drew more of their income from selling and leasing land, from selling livestock, and from federal rations.
- They drew less from “native industries”, which covers traditional harvesting and art sales, and less from wages.
- Incomes in the first decades after the slaughter were held up by selling and leasing capital assets.

Net In-Migration Per 100 People



Exposed to Slaughter



Not Bison Reliant

Figure 7: net in-migration

- Net in-migration, 1940 to 2000. Neither group shows net out-migration in any decade.
- In-migration is systematically larger in the regions that were never bison-reliant.
- Mobility was restricted by law until 1924, and confirmed only by the Nationality Act of 1940.
- From 1954 the federal government actively encouraged relocation to cities. Many who went experienced hardship and returned.

Table 6: was it the able who left?

- Members of bison nations are less likely to hold a bachelor's degree.
- Proportionately more of those who do hold one live on Native homelands, which points to retention rather than flight.
- They are no more likely to have stayed in the same house, and no more likely to have moved across state lines.
- Selective out-migration does not generate the income gap.

Tables 7 to 9: credit and recovery

- Exposed reservations with more banks in 1980 gained more net in-migration over the following decade.
- Exposed places farther from a bank in 1870 gained less. The same pattern repeats for 1990 to 2000.
- Bison nations farther from historical banks are still poorer today, and historical population density does not account for it.
- Bison nations that lost more banks between 1929 and 1936 do worse still. Shocks amplify one another.

The argument

- A resource loss compressed into a decade removed the productive base of the wealthiest societies on the continent.
- Restrictions on movement and on occupation closed the usual margins of adjustment.
- Thin credit closed the remaining one, and left the region unattractive to both capital and people.
- Some Native communities today are still 50 to 70 miles from the nearest bank.

Comparisons

- The authors place this alongside American coal and steel in the 1980s, the North Atlantic cod collapse of the 1990s, the Dust Bowl, and manufacturing after 2000.
- Brazil: the Amazon rubber boom collapsed after Southeast Asian plantations took the market. Manaus and Belém never regained their position.
- For discussion: which ingredient does the work here, the size of the shock or the absence of credit and mobility?

Forced coexistence

Dippel (2014)

FORCED COEXISTENCE AND ECONOMIC DEVELOPMENT: EVIDENCE FROM NATIVE AMERICAN RESERVATIONS

BY CHRISTIAN DIPPEL¹

Studying Native American reservations, and their historical formation, I find that their forced integration of autonomous polities into a system of shared governance had large negative long-run consequences, even though the affected people were ethnically and linguistically homogenous. Reservations that combined multiple sub-tribal bands when they were formed are 30% poorer today, even when conditioning on pre-reservation political traditions. The results hold with tribe fixed effects, identifying only off within-tribe variation across reservations. I also provide estimates from an instrumental variable strategy based on historical mining rushes that led to exogenously more centralized reservations. Data on the timing of economic divergence and on contemporary political conflict suggest that the primary mechanism runs from persistent social divisions through the quality of local governance to the local economic environment.

KEYWORDS: Economic development, population heterogeneity, political centralization, institutions, factional politics.

The question

- Artificial jurisdictions, whose borders do not match what the people inside them wanted, tend to perform badly.
- The usual explanation is ethnic and linguistic division.
- Dippel isolates a different thing: people with **no history of shared governance** placed under one government.
- Reservations allow the separation, because the affected people are ethnically and linguistically the same.

How reservations were formed

- The first reservations date from the 1850s. The overwhelming majority of present-day reservations were formed by the 1890s.
- The government's aim was to free up land for settlement and to monitor Native Americans once they were on it.
- Different tribes were kept apart, because combining them was likely to produce open violence. So each reservation maps to one tribe.
- A tribe was not a government. Decisions were made at the level of the local band.

“the notion of tribe in Apachean cultures is very weakly developed. Essentially it was only a recognition that one owed a modicum of hospitality to those of the same speech, dress, and customs.”

Opler (1983), quoted by Dippel



FIGURE 1.—U.S. reservation centroids.

The sample

- The 2000 census reports averages for 210 reservations with more than 100 residents.
- Dropped: 12 that are not federally recognized, 9 founded after the 1934 Indian Reorganization Act, 7 that combine more than one tribe.
- What remains: **182 reservations** across 78 tribes, covering over 90 percent of the on-reservation Native American population.
- Outcome: income per capita on the reservation.

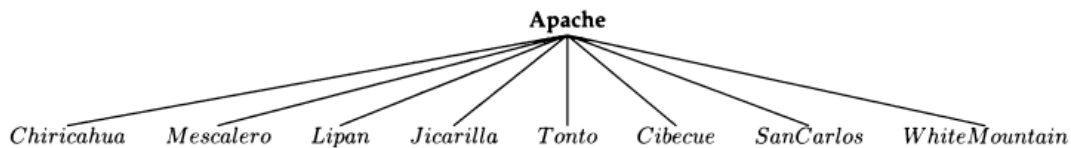


FIGURE 2.—The bands, that is, local communities, of the Apache tribe.

The measure

- Forced coexistence: the reservation was formed out of more than one band.
- The Apache had eight bands. Chiricahua, Mescalero and Lipan went to Mescalero. Tonto, Cibecue, San Carlos and White Mountain went to San Carlos.
- Only the Jicarilla obtained a reservation of their own.
- Mescalero and San Carlos are coded as forced coexistence. Jicarilla is not.

The confound

- Centralized tribes were more likely to end up on combined reservations, either by negotiating as a tribe or because the government preferred it that way.
- Historical centralization comes from Murdock's *Ethnographic Atlas*: whether political authority existed above the local community.
- Every North American tribe is coded 0, 1 or 2, and 57 of the 78 tribes are coded 0.
- Heterogeneity and prior state-building pull in opposite directions, so both have to be in the comparison.

TABLE I
JOINT DISTRIBUTION OF FORCED COEXISTENCE AND HISTORICAL CENTRALIZATION^a

	Forced Coexistence = 0		Forced Coexistence = 1		Total
	Number	log(p.c. income)	Number	log(p.c. income)	
Historical centralization = 0	66	9.34 (0.41)	78	8.98 (0.27)	144
Historical centralization = 1	3	9.58 (0.39)	35	9.26 (0.32)	38
Total	67		115		182

^aThe total number of reservations is 182. For each of the four cells, the table reports the number of reservations as well as the mean and standard deviation (in parentheses) of log(per capita income). If the data are broken down by tribe instead of by reservation, 57 tribes have a level of historical centralization of 0 and 21 of 1. (There are 78 tribes in the data.)

Table 1: the raw pattern

- Of the 38 reservations belonging to historically centralized tribes, 35 were formed above the level of the band.
- The 78 combined reservations of non-centralized tribes have clearly lower income per capita.
- The 35 combined reservations of centralized tribes do not.
- Read without holding centralization fixed, this table gives the wrong answer.

TABLE II
BALANCING TESTS^a

Dependent (Below)	Regressors					
	Forced Coexistence Only			Forced Coexistence, Conditional on Historical Centralization		
	Coeff.	<i>t</i> -Stat.	<i>R</i> ²	Coeff.	<i>t</i> -Stat.	<i>R</i> ²
<i>Panel A: Reservation Characteristics</i>						
Surround. p.c. income	-0.051	(-1.417)	0.024	-0.042	(-0.986)	0.030
Surround. p.c. unempl.-rate	-0.028	(-0.646)	0.003	-0.003	(-0.046)	0.025
Distance to major city	0.083	(0.321)	0.002	0.072	(0.260)	0.002
log(Ruggedness, Reserv.)	-0.140	(-0.646)	0.003	0.086	(0.427)	0.066
log(Re-Area in sqkm)	0.182	(0.291)	0.001	0.218	(0.292)	0.001
<i>Panel B: Tribe Characteristics</i>						
Historical centralization	0.266***	(2.973)	0.101	—	—	—
Percent calories from agriculture	-1.021	(-0.737)	0.037	-1.456	(-1.059)	0.096
Sedentariness	-0.290	(-0.323)	0.005	-0.616	(-0.699)	0.062
Complexity of local community	0.075	(0.657)	0.006	0.013	(0.095)	0.041
<i>D</i> (Wealth distinctions)	0.211	(1.469)	0.025	0.318	(1.601)	0.080
<i>Panel C: Endogenous Reservation Controls</i>						
log(Population)	0.501*	(1.896)	0.038	0.450	(1.417)	0.041
Pop-Share Adult (0-100)	-3.495***	(-2.929)	0.086	-3.781***	(-2.944)	0.092
<i>D</i> (Casino)	0.029	(0.319)	0.001	0.008	(0.085)	0.005

^aEach row reports on a separate regression of one control variable on forced coexistence. Each row reports the same regression in two different specifications: first, with forced coexistence as the only regressor, then additionally controlling for historical centralization. In both specifications, I report only the coefficient on forced coexistence. The number of observations is 182 in each regression. The *t*-statistics are for standard errors that are two-way clustered at the tribe and the state level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table II: what is balanced

- Reservation size, ruggedness, distance to a major city and surrounding unemployment look balanced.
- Income in the surrounding counties is somewhat lower where there was forced coexistence, though not sharply.
- Among tribe characteristics, historical centralization is the one that moves strongly with forced coexistence.
- Population size and the adult share are less balanced, and are plausibly consequences rather than causes.

Three ways of asking

- Compare reservations holding the observable characteristics fixed.
- Compare reservations of the same tribe with each other, which removes whatever the tribe carries with it.
- Use variation the bands themselves did not control: gold and silver rushes on their ancestral homelands.
- The three rest on different assumptions and give the same answer: forced coexistence, about **30 percent** lower income today.

Where the mining variation comes from

- Where the government judged Native land more valuable, it formed fewer and more concentrated reservations, to free up land and to keep tribes from returning to their homelands.
- Native North Americans, unlike the Inca, did not have mining. Where gold and silver lay was a matter of geology.
- Mining maps from the 1880 census, laid over digitized maps of ancestral homelands, band by band.
- The Black Hills gold rush of the 1870s happened on Dakota Sioux territory, militarily the strongest tribe on the Plains. Strength was no deterrent.



FIGURE 3.—Geo-database of gold (triangle) and silver (circle) mining clusters.



FIGURE 4.—Mescalero Reservation (in black) with surrounding counties (shaded), ancestral homelands (in white), and gold (triangle) and silver (circle) mining clusters.

Keeping the two apart

- The mining that counts is on the ancestral homeland, which is spatially separate from the reservation and from its present-day economic environment.
- Mining in the surrounding counties enters separately as a control.
- Distance between reservation and homeland controls for having been removed farther.
- Boas heights test whether reservation formation was more violent where there was mining: growing up during formation did reduce adult height, but no more so where mining was present.

The magnitudes

- A mining rush, meaning one standard deviation more mining value, raises the probability of forced coexistence by 6 to 11 percentage points.
- Both gold and silver reduce income on the reservation today.
- Estimates that use mining are larger than the plain comparison.
- Dippel's reading: the bands that combined only under the extra pressure of mining were the ones with the weakest prior ties to each other.

When the gap opened

- Until the late 1980s, reservations were still run in practice by the BIA, whatever powers the 1934 IRA gave tribal councils on paper.
- From the late 1980s, federal funding tightened and money arrived as block grants. Reservation governments became real governments.
- Prediction: forced coexistence should start to matter at that point, and not before.
- In the income panel from 1970, no clear differential effect in 1980 or in 1990. The gap appears in **2000**.

Politics in the newspapers

- ProQuest searches for reports of internal conflict and of corruption on each reservation, 1981 to 1990 and 1991 to 2000.
- Total coverage of each reservation is held fixed, so this is not simply a count of how often a place is in the news.
- Forced coexistence raises reported internal conflict, and more so when the conflict explicitly involves the reservation government.
- The conflict results appear mainly in the 1990s, matching the timing of the income divergence. Corruption is less robust once all controls go in.

Why local conflict costs income

- Under the IRA constitutions, tribal councils hold executive, legislative and judicial power together, including control over reservation courts and their judges.
- Public goods come largely from the BIA, so under-provision of public goods is unlikely to be the channel.
- Commercial disputes are settled in reservation courts, so factional politics turns into contracting uncertainty. Haddock and Miller call this the **sovereign's paradox**.

Table VIII: where the income difference sits

- Forced coexistence lowers wage income per capita, average salary, and the share employed.
- It does not lower transfer income, and BIA program spending is if anything slightly higher.
- No sign of weaker integration with the surrounding economy: same likelihood of working outside the county, same travel time, same share speaking English at home.
- The difference is in income earned on the reservation.

Discretion in the allotment era

Dippel, Frye and Leonard (2024)

Bureaucratic discretion in policy implementation: evidence from the Allotment Era

Christian Dippel¹ · Dustin Frye²  · Bryan Leonard³

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Abstract

From 1887 to 1934, the federal government broke up millions of acres of tribally owned reservation lands and allotted them to individual Native American households. The Bureau of Indian Affairs' local "Indian agents" oversaw this highly contingent conveyance of property rights. They initially managed the allotted land held in trust, and then later decided when and if to re-title it to fee simple. Building on and going beyond the literature showing that bureaucratic incentives matter greatly for policy implementation, our paper studies empirically to what extent individual agents' idiosyncratic preferences and discretion shaped this process. We find that individual agents were statistically important drivers of policy implementation, introducing an element of historical randomness into the legacy of allotment, which continues to shape the distribution of land titles on reservations to the present day.

The question

- Between the 1887 Dawes Act and the 1934 IRA, the federal government broke up millions of acres of tribal land into individual allotments.
- The rights conveyed were contingent. Allotments started in trust, and only a local BIA agent's declaration of **competency** turned them into fee simple.
- The question: would two otherwise identical allotments have had different fates under different agents?

Four categories of land

- Before allotment: tribal land, held in trust by the federal government.
- Allotment created individual parcels, which began as *allotted trust*: not sellable, not usable as collateral, not written into wills.
- *Fee simple* was the next step: full individual ownership, alienable.
- Land left over after every household had been allotted was declared *surplus* and sold to settlers.

Bootleggers and Baptists

- Reformers wanted private ownership as the route to assimilation. “Kill the Indian to save the man.”
- Local politicians and land speculators wanted reservations broken up so the land could be bought.
- The federal government wanted the “Indian question” resolved, and required that proceeds from surplus sales be held in trust for “education and civilization”.
- All three favored breaking the reservations into individual allotments.

“The General Allotment Act is a mighty pulverizing engine to break up the tribal mass. It acts directly upon the family and the individual.”

Theodore Roosevelt, first annual message to Congress, December 1901

The sequence

- **1887**: Grover Cleveland signs the General Allotment Act. Title held in trust for 25 years, or until the local agent declares competency.
- **1906**: the Burke Act speeds up the issuing of allotments.
- **1913 to 1916**: Franklin Lane's Competency Commission tours the reservations to speed up conversion, outside the usual BIA chain. Over 20,000 patents, roughly 1 million acres.
- **1924**: the Indian Citizenship Act ends the rule tying fee simple to citizenship. 1928: the Meriam Report.
- **1934**: the IRA ends allotment and freezes the map of titles as it then stood.

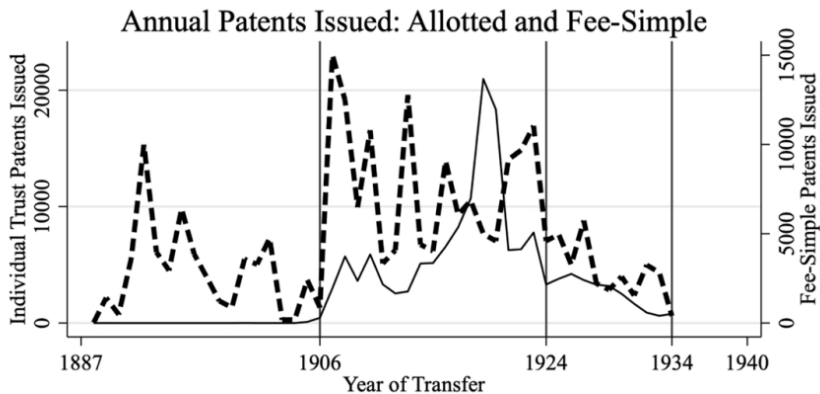


Fig. 2 Aggregate flow of allotments and transfers into fee simple. This figure tracks the flow of trust patents issued (dashed line) and the flow of patents subsequently transferred into fee simple in the BLM data (solid line), described in Sect. 3. The Dawes Act officially began in 1887, the Burke Act was introduced in 1906, and the IRA passed in 1934

Figure 2: two different flows

- Dashed line: trust patents issued. Solid line: allotments later converted to fee simple.
- The Burke Act of 1906 accelerates allotment. It does not accelerate conversion.
- Conversion peaks during the First World War years.
- Conversion slows after 1920 and new allotments after 1924. The IRA stops both in 1934.

Why incompleteness matters

- Had the policy run its full course, every eligible acre would have been allotted and then converted, and the agents' discretion would have washed out.
- It did not run its course. Every reservation reached 1934 at a different point in the process.
- The IRA kept allotted trust in trust, kept fee simple as fee simple, and left remaining tribal land tribal.
- Local decisions taken before 1934 therefore still shape the distribution of titles on reservations today.

What the agents did

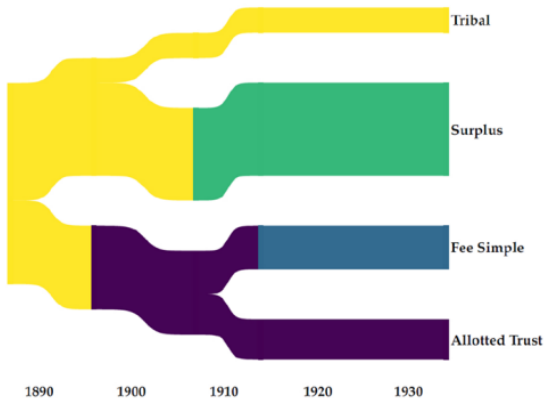
- BIA agents ran schools, pressed adults to work, managed finances, reported to Washington, and handled land.
- For fee conversion they were instructed to assess English-speaking ability, literacy, self-sufficiency and other perceived signs of assimilation.
- The Dawes era straddles the patronage system and a professionalized federal bureaucracy, with relatively little local supervision.
- Meriam Report on the process: “The original allotments of land to the Indians were generally made more or less mechanically.”

Why agents might have differed

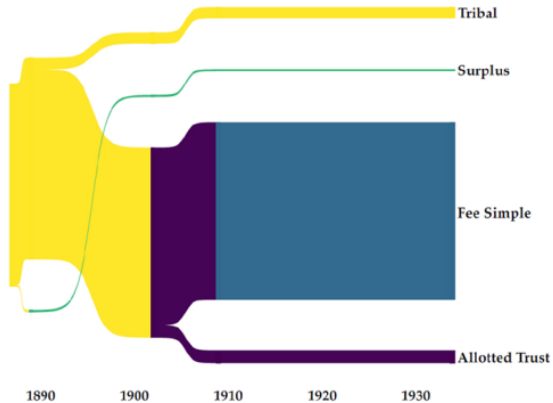
- The BIA's own annual reports describe three types: those who wanted tribal ownership preserved, those who wanted the land opened at once, and those who feared allottees were not ready.
- Local interests had money at stake, because fee simple land could be bought.
- Banner: "there was considerable room for fraud in these transactions, partly because of the ignorance of many of the sellers, but largely because government officials and land purchasers could find it easy and profitable to work together."
- The paper does not start from a motive. It first asks whether agent identity mattered at all.

The data

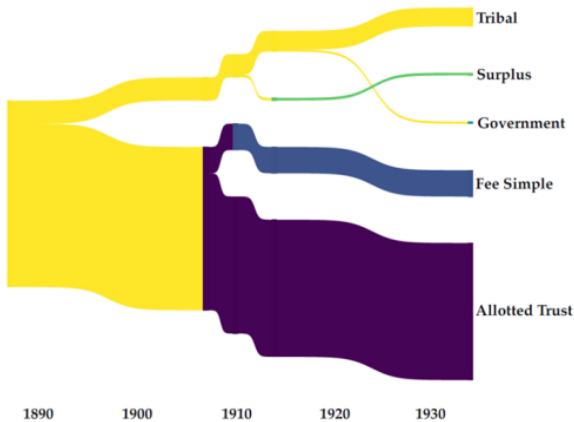
- Land: every patent filed with the General Land Office, digitized by the Bureau of Land Management, mapped onto the Public Land Survey System.
- Each 160-acre quarter section gets a full history of the titles it was held under, with dates.
- Agents: a reservation-year panel from 1879 to 1940, from Interior Department employment rosters, BIA annual reports and the Indian Census Rolls.
- Agencies were renamed, split, merged and terminated, so the panel needs crosswalks from reservation to agency to agent.
- Outcome: for each reservation and year, the share of its allotments already converted to fee simple.



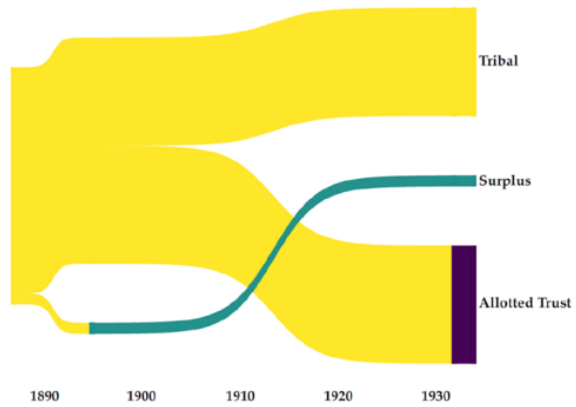
(a) Fond du Lac



(b) White Earth



(c) Crow



(d) Northern Cheyenne

Figure 3: four reservations

- The thickness of each band is the share of land in that category. The color changes mark the dates of transition.
- Fond du Lac: about 44 percent allotted, about 48 percent declared surplus, and roughly half of the allotted land converted.
- White Earth: nearly the whole reservation allotted, and nearly all of it converted to fee simple.
- Crow: most land allotted, very little converted. Northern Cheyenne: allotted so late that none of it converted before 1934.

What Figure 3 does not show

- Implementation was clearly not uniform across reservations.
- But the differences in initial allotment are reservation-level: land quality, distance to mining, population pressure, land available as surplus, proximity to railroads.
- Those decisions are studied elsewhere. This paper takes the next step only, conversion from allotted trust to fee simple.

Fig. 4 Regression of share allotments converted to fee. This figure plots the year fixed effects from a regression of reservation r 's share of allotments that have been converted to fee simple on only reservation and year fixed effects

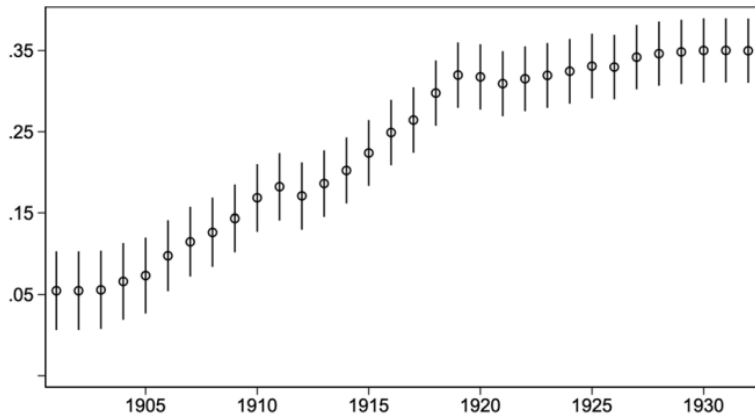


Figure 4: the average reservation

- In 1900, about 5 percent of allotments on the average reservation had been converted to fee simple.
- Conversion accelerates visibly after 1907, following the Burke Act.
- By 1915, about 20 percent.
- It begins to plateau in the early 1920s at about 30 percent.

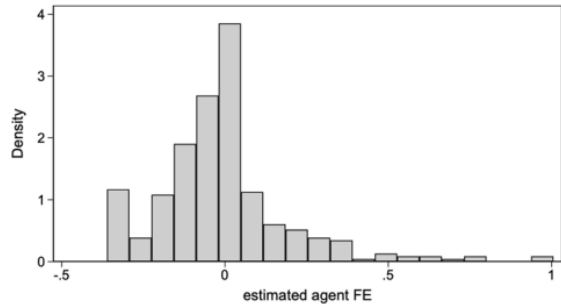
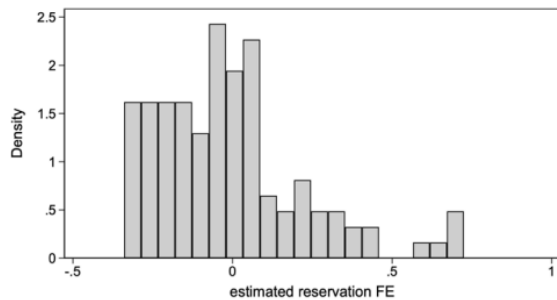


Fig. 5 Comparing reservation and agent fixed effects. This figure's left panel displays the distribution of 116 reservation fixed effects, while the right panel displays a (separately estimated) distribution of 369 agent fixed effects

Figure 5: two distributions

- Left panel: 116 reservations, and they differ a great deal in how far conversion went.
- Right panel: 369 agents, and some are clear outliers in their propensity to convert.
- This is suggestive only. An agent posted during a period favorable to conversion would look like a converting agent.



Fig. 1 RIFLE permutations illustrated. This figure shows the principle behind RIFLE's permutation tests. While three permutations are shown, $4! = 24$ are possible with four tenure-spells

The shuffling test

- A reservation's history is a sequence of agent spells.
- **RIFLE** reshuffles the order of those spells inside the reservation, holding fixed how many agents there were and how long each stayed.
- Only the order changes. The average reservation had nearly nine agents, so there are hundreds of thousands of possible orders.
- The question: does the true order explain the reservation's conversion series better than counterfactual orders do?

Fig. 6 Baseline RIFLE results.

This table displays the distribution of adjusted R-squared values that arises from permuting the agent fixed effects a large number of times. Under the hypothesis that agents mattered, the R-squared resulting from the permuted data should be smaller than the R-squared using the true data. If agents had no discretionary power, on the other hand, the R-squared resulting from the real data (dashed vertical line) should be centered on the distribution of R-squared values

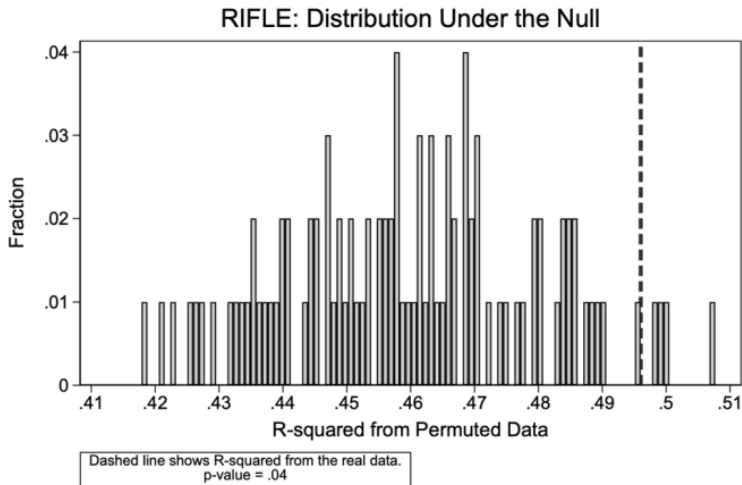


Figure 6: baseline result

- The histogram is the fit obtained from shuffled orders. The dashed line is the fit from the true order.
- Only 4 percent of shuffles do better than the truth.
- Whether an allotment became fee simple before 1934 depended on who happened to be running the reservation.

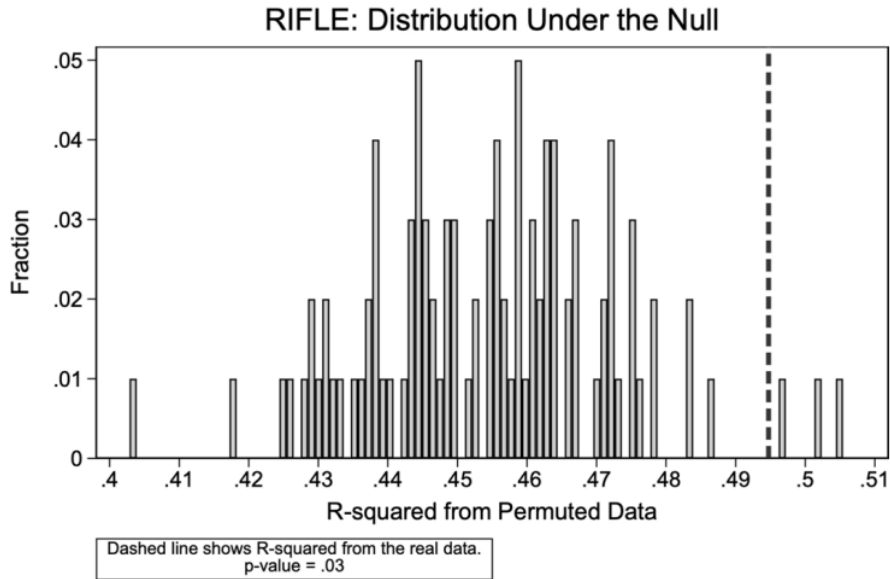


Fig. 7 RIFLE results with the Competency Commission

Figure 7: the Competency Commission

- Treat the commission as the agent in charge during the years it was on a reservation, in place of the resident agent.
- The true order now stands further out: 3 percent of shuffles beat it, against 4 percent before.
- The commission was created in Washington precisely to accelerate conversion, and it mattered over and above the local agents.

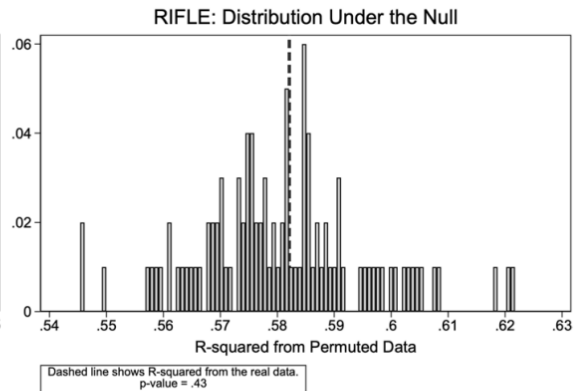
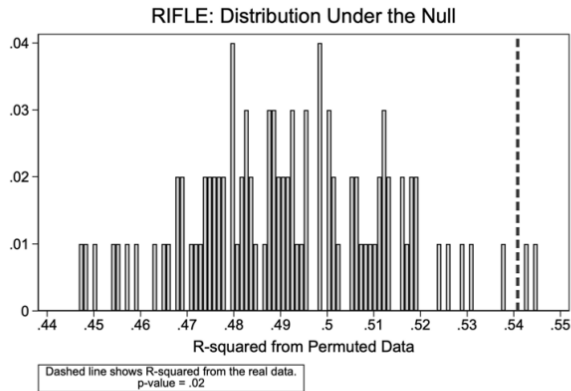


Fig. 8 The importance of bureaucratic discretion with and without clear policy guidance. This figure's left panel displays the RIFLE results for the time window 1887–1906 plus 1925–1935, i.e., excluding the era from the Burke Act to the Indian Citizenship Act during which federal guidance on Indian allotment was the clearest and most cohesive. The right panel displays the RIFLE results for the time window 1907–1924

Figure 8: discretion and policy guidance

- Left panel: 1887 to 1906 plus 1925 to 1935, when federal guidance was neither clear nor cohesive. 2 percent of shuffles beat the true order.
- Right panel: 1907 to 1924, between the Burke Act and the Indian Citizenship Act, when guidance was clearest. 43 percent beat it, and the true order sits in the middle of the distribution.
- Agent discretion mattered where the instruction from above was vague, and not where it was clear.

What the test cannot say

- It says agents differed from one another. It does not say whether agents on average sped conversion up or slowed it down.
- It does not separate an agent's own convictions about fee simple from an agent's receptiveness to local pressure or payment.
- It says nothing about whether converting a given parcel was good or bad for the allottee.
- The paper then looks for evidence compatible with rent-seeking.

Fig. 9 BIA agents' real wages and the Consumer Price Index (CPI). This figure shows the decline in agents' real wages as a result of the sharp increase in the consumer price index in the wake of World War I. Income data from United States Government Printing Office (1934)

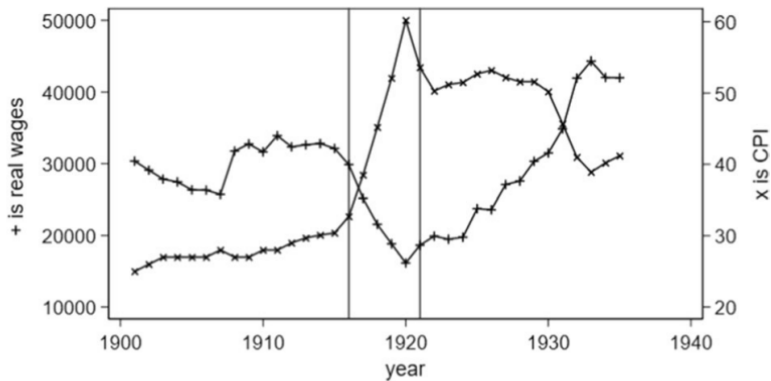


Figure 9: agents' real wages

- Nominal salaries were rigid. The consumer price index jumped in the wake of the First World War.
- Real pay fell sharply between 1916 and 1920.
- That window coincides with the peak of conversion to fee simple in Figure 2.

Table 1: real wages and conversion

- Outcome: the share of a reservation's allotments converted to fee simple.
- A 1 percent fall in the agent's real wage goes with a 4 to 7 percent rise in the rate of conversion.
- The pattern holds with reservation, agent and year all held fixed.
- Compatible with rent-seeking, and not a full account of motives. Settlers, speculators and local business stood to gain from more land in fee simple.

The three papers together

- Feir, Gillezeau and Jones: a resource shock the affected nations could not absorb, because movement and credit were closed to them.
- Dippel: reservation boundaries drawn above the level of the band, still worth 30 percent of income a century and a half later.
- Dippel, Frye and Leonard: within that, which acres ended in whose hands turned partly on which official happened to be posted where.
- All three describe institutions designed from outside for populations with no say in them, and then frozen in place by later law.